



Construction of physically crosslinked cellulose nanofibrils/alkali lignin/montmorillonite/polyvinyl alcohol network hydrogel and its application in methylene blue removal

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Abstract Herein, cellulose nanofibrils (CNF), alkali lignin (AL), and montmorillonite (MMT) were used to produce reinforced polyvinyl alcohol (PVA) hydrogels. The effects of MMT and AL contents on the rheological properties of reinforced hydrogel were studied. Compared with PVA/CNF hydrogel, the storage modulus of 40 wt% MMT-reinforced PVA hydrogel was increased by 41.4%. The rheological properties of MMT-enhanced PVA hydrogel could be adjusted by the variation of MMT loading. Also, as the PVA matrix had a synergistic effect with the

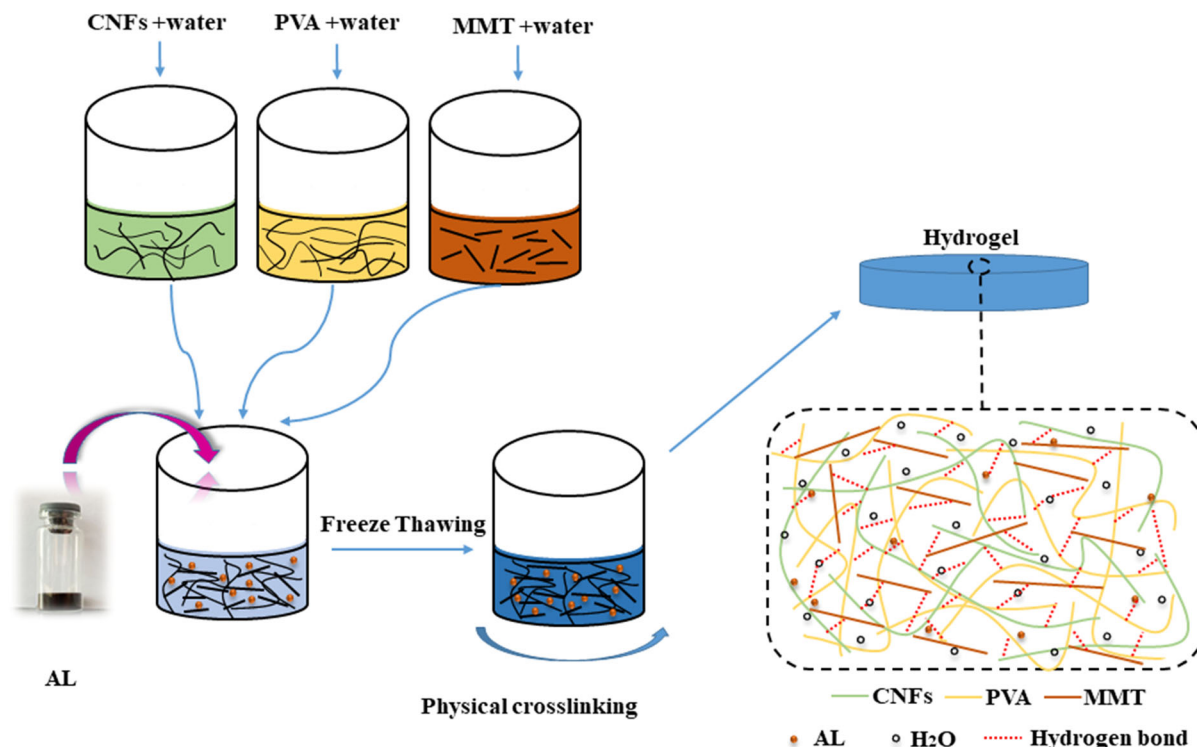
embedded MMT and AL, the composite hydrogel demonstrated high efficiency in the removal of methylene blue dye (MB) from wastewater. Adsorption tests conducted at various time intervals (60–360 min) show that the hydrogels containing same content of MMT had higher removal efficiency. The MB adsorption of PVA/2CNF-0Li-40MMT was over 98.0%, whereas its adsorption equilibrium time and maximum adsorption capacity (q_m) were 360 min and 67.2 mg/g, respectively. However, an extremely high content of MMT reduced the MB adsorption rate.

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Graphic abstract



Keywords Hybrid hydrogel · Alkali lignin · Montmorillonite · Dye adsorption

Introduction

Industrial production of materials such as paper, textiles, and food mostly produces wastewater containing dyes (Schwarzenbach et al. 2006). These pollutants accumulate in the form of ions or compounds in the ecosystem, causing severe harm to human health, as well as health of aquatic animals and plants (Shannon et al. 2008). The current methods for the removal of dye-containing wastewater include adsorption (Deng et al. 2020), biodegradation (Kurniawan et al. 2020), flocculation (Zolezzi et al. 2018), and photocatalytic degradation (Reddy et al. 2016). Owing to several advantages such as high treatment efficiency and reusability, the adsorption method is one of the important methods for the removal of dyes from wastewater. Adsorption material is the key

determinant of efficiency and environmental protection of dyes removed via adsorption.

In recent years, cellulose nanofibrils (CNF) with a three-dimensional structure has displayed multiple advantages such as high porosity, large specific surface area, excellent reproducibility, low density etc. (Carpenter et al. 2015; Nascimento et al. 2018). These properties endow CNF with application in the wastewater dye removal field (Shen et al. 2019). The chain structure of CNF was employed to improve the mechanical strength of ionic conductive polyvinyl alcohol (PVA) hydrogel (Nascimento et al. 2018; Sapalidis 2020), in order to develop a potential solution for the dye removal from sewage. In the last few years, several studies have focused on the use of lignin as a starting material for the production of hydrogels and aerogels for toxic compound removal (Grishechko et al. 2013; Penaranda and Sabino 2010). We had successfully prepared physically cross-linked hydrogels containing alkali lignin (AL) for dye adsorption. However, the presence of AL has not provided sufficient mechanical strength to hydrogels.

These prepared hydrogels collapsed after swelling that rendered them unsuitable as dye adsorbents.

In general, it is recommended to add fillers for the enhancement of mechanical properties of hydrogels. Using chemically and physically crosslinking networks, these fillers can cooperate to improve the mechanical properties of the hydrogels (Sun et al. 2012; Tao et al. 2020). As most fillers to these hydrogels are usually expensive and toxic, the abundant and cheap AL was explored as a hydrogel filler for the dye removal to increase the economic and environmental benefits. The hydrogel was combined with montmorillonite (MMT) and PVA to formulate a composite material as the environmentally-stable carrier for nanoparticles. Incorporation of MMT is a useful method to improve the performance of PVA-based hydrogels. It has great potential in the production of nanocomposites that possess high adsorption capacity and other superb physical properties (Lv et al. 2017; Nguyen et al. 2020). A Lignin-based hydrogel with improved mechanical strength has been prepared previously (Bian et al. 2018). However, only a few works have discussed the effect of AL and MMT contents on the mechanical strength of hydrogels. Herein, strong and complex hydrogels of PVA/CNF/AL/MMT (PCAM) were prepared for the water-soluble dye pollutant removal. Also, the dye adsorption performance and mechanism of PCAM constituents were investigated.

Materials and methods

Materials

Cellulose nanofibrils (CNF) were produced from bleached dry hardwood pulp. Polyvinyl alcohol (PVA, MW = 125,000 g/mol, 99% hydrolyzed), montmorillonite (MMT), and alkali lignin (AL, MW = 10,000 g/mol) were purchased from Sigma-Aldrich (USA).

Preparation of hybrid hydrogels

Montmorillonite (3 g) was dispersed in 100 mL of deionized water under vigorous stirring to obtain 3.0 wt% montmorillonite suspension. The CNF suspension (0.75 wt%) was prepared from the wood pulp as per the method reported by Zhang et al. (Zhang et al.

2016). The hybrid aerogels were formulated by mixing CNF, AL, MMT, and PVA. The CNF and PVA loadings were fixed at 2 wt% and 58 wt%, respectively, whereas the AL loading was varied at 0, 10, 20, 30, and 40 wt%; while the remaining portion was MMT. The control aerogel was prepared by mixing CNF (2 wt%) with PVA (98 wt%). Total content of mixture in all the hydrogels was maintained at 4 wt% (Table 1).

Characterization

The structural features of CNF were determined via atomic force microscopy (AFM, Germany). The CNF suspensions were diluted to 0.1%, and then, used to obtain the images of individual cellulose fibers. The morphology of PCAM hydrogels was observed under scanning electron microscope (SEM, USA). The dried hydrogels were sputtered with gold powder to improve their conductivity. IR spectra of the samples were measured using Fourier transform infrared spectrometer (FTIR, Germany) in the wavenumber range of 4000 to 600 cm^{-1} at the resolution of 0.5 cm^{-1} . Thermogravimetric analysis (TGA, Germany) of the hydrogel samples was conducted using a thermogravimetric analyzer in a stream of pure nitrogen flow at the rate of 20 mL/min. Hydrogel samples (10 mg) were heated from 100 to 600 °C. X-ray spectra of the hydrogel samples were measured using an Ultima IV diffractometer (XRD, Japan) in the 2θ range of 5° to 35°. The crystallinity index (CrI) was estimated from Eq. (1) using the Segal method (Segal et al. 1959)

$$CrI = \frac{I_{200} - I_{am}}{I_{200}} \times 100 \quad (1)$$

where I_{200} is the maximum peak intensity at 2θ angle of 22.6° and I_{am} is the intensity at 2θ angle of 18.6°.

Each hydrogel sample was removed from the culture dishes (15 × 11 mm), and then, dried at 105 °C. Water content (W_1) in the hydrogel samples was determined from mass decrease using Eq. (2) (Cha et al. 2012):

$$W_1 = \frac{m}{M} \times 100 \quad (2)$$

where M is the initial weight (g) of hydrogel samples and m is the weight of the hydrogel samples after drying at the temperature of 105 °C.

Table 1 PCAM hybrid layered nanocomposite hydrogels

Sample	PVA: CNF:AL: MMT wt %	Mass ratio of PVA: CNF:AL: MMT
PVA/2CNF	98/2	1:0.02
PVA/2CNF-0Li-40MMT	58/2/0/40	1:0.03:0:0.69
PVA/2CNF-10Li-30MMT	58/2/10/30	1:0.03:0.17:0.52
PVA/2CNF-20Li-20MMT	58/2/20/20	1:0.03:0.34:0.34
PVA/2CNF-30Li-10MMT	58/2/30/10	1:0.03:0.5:0.17
PVA/2CNF-40Li-0MMT	58/2/40/0	1:0.03:0.69:0

Swelling ratio of the samples was determined in a 10 mL vial by covering the vacuum-dried samples with a mass of about 40 mg. Samples were placed at room temperature for 24 h. Thereafter, the samples were removed from the vial and quickly dried with paper before the weight determination of swollen samples. Swelling ratio was calculated using Eq. (3)

$$\text{Swelling Ratio} = \frac{M_s - M_i}{M_s} \quad (3)$$

where M_i and M_s are the initial and swollen mass, respectively.

The rheological behavior of hydrogel samples was evaluated using an RS6000 rheometer (HAAKE, Germany) equipped with a Platte P20 TiL and cone plate P20 TiL. The sample was cut into cylindrical shape with the diameter of 20 mm and thickness of 1 mm. The modulus was recorded in the linear viscoelastic zone wherein the modulus is independent of the applied strain (Liu et al. 2016).

Methylene blue (MB) adsorption

The hydrogel samples (30 mg, oven-dried) were placed in a 50 mL beaker containing 20 mL of MB solution (0.25 mg/mL). The beaker was placed in a water bath at 30 °C and 50 °C. The mixture system continually oscillated at 80 rpm for 360 min. Concentration of MB in the supernatant was determined using a spectrometer (UV-1800). The adsorption ability at different time intervals was calculated using Eq. (4)

$$Q_t = \frac{V(C_0 - C_e)}{m} \quad (4)$$

Removal efficiency (R) was obtained using the following equation:

$$R = \frac{(C_0 - C_e)}{C_0} \times 100\% \quad (5)$$

where C_0 and C_e are the initial and equilibrium MB concentrations in the solution (mg/mL), respectively. V (mL) and m (g) are the total volume of solution and weight of the hydrogel samples (oven-dried), respectively. All the tests were performed in duplicate.

Results and discussion

Preparation and characterization of PCAM hydrogels

TEMPO (2,2,6,6-tetramethylpyperidine-1-oxyl) oxidation and ultrasonic dispersion were used to produce CNF from the bleached dry hardwood pulp. The yield of CNF obtained from TEMPO oxidation and defibrillation process was 89.9%. The reduction of yield was primarily due to the loss of hemicellulose and amorphous cellulose. The FTIR spectra of PCAM hydrogels displayed the corresponding characteristic peaks of the employed materials. Both the spectra exhibited same peaks: (1) stretching and bending vibrations of $-\text{OH}$ at 3400 cm^{-1} and 1630 cm^{-1} , respectively; (2) $\text{C}-\text{O}$ stretching vibration at 1038 cm^{-1} ; and (3) the $-\text{CH}_2$ and $-\text{CH}_3$ stretching vibrations at 2939 and 2870 cm^{-1} , respectively. The presence of AL can be proven by the skeleton extension of the aromatic ring at 1600 and 1505 cm^{-1} . The peaks at 1460 cm^{-1} and 1260 cm^{-1} corresponded to the benzene ($\text{C}=\text{C}$) and aromatic ring ($\text{C}=\text{O}$), respectively. The AL peaks were similar to those observed in the softwood prepared by a commercial alkali pulping process (Hu et al. 2016; Hu and Lo, 2013). As shown in Fig. 1a, the peaks at

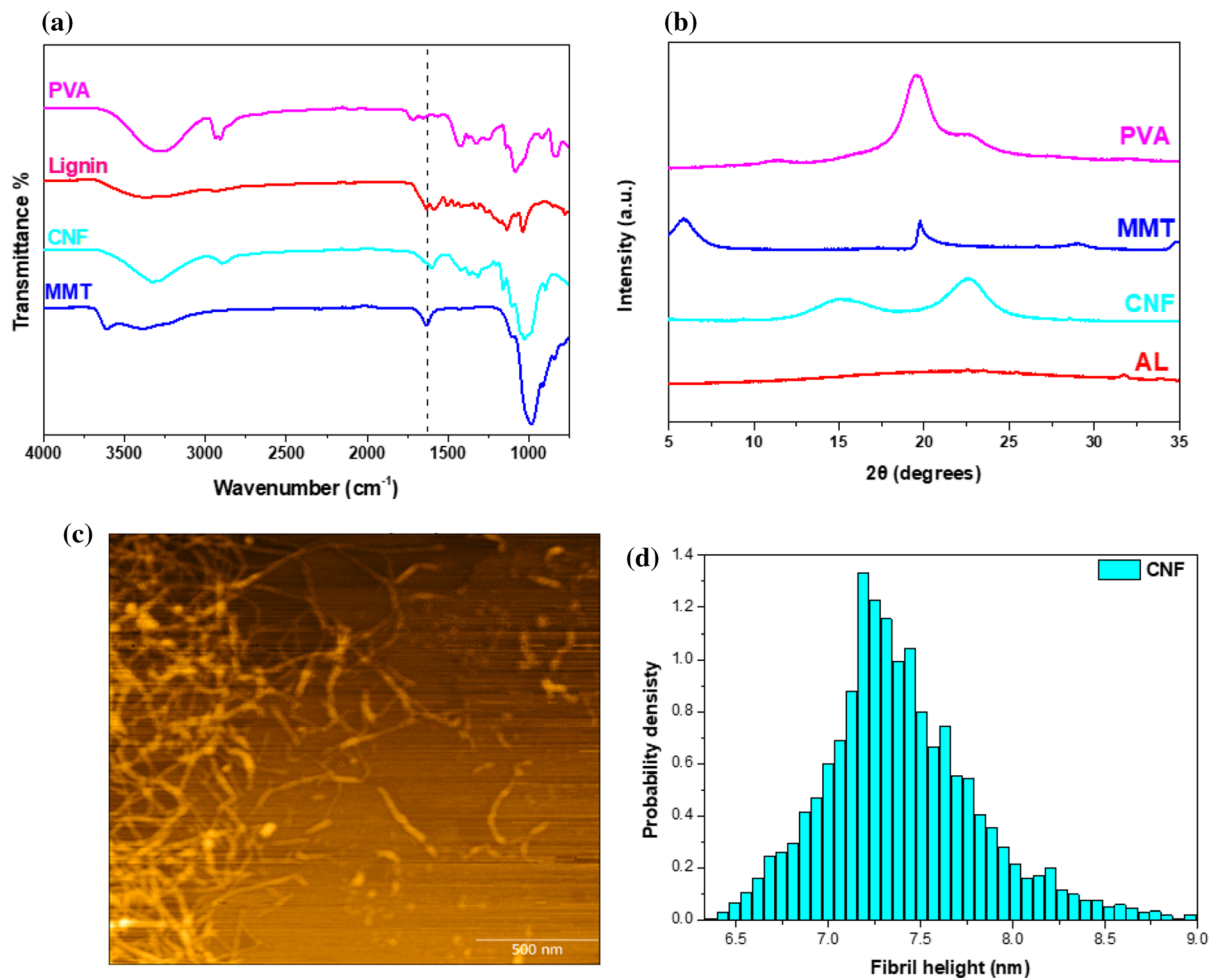


Fig. 1 **a** FTIR spectra of cellulose nanofibrils (CNF), alkali lignin (AL), montmorillonite (MMT), and polyvinyl alcohol (PVA); **b** XRD patterns of CNF, AL, MMT, and PVA. **c** AFM image (Scale bar = 1 μ m) of CNF; **d** Height distribution shown from AFM image

Table 2 Physical, rheological, and thermodynamic properties of hybrid hydrogels at different AL or MMT loadings

Samples	Moisture content (wt%)	Swelling ratio (%)	Maximum G' with LVR, G' max (Pa)	Maximum G'' with LVR, G'' max (Pa)	T _{max} (°C)	Weight loss at 600 °C
PVA/2CNF	94.2 $\pm$ 1.2	621.6 $\pm$ 22.3	1717.8 $\pm$ 127.1	90.5 $\pm$ 8.6	287.6 $\pm$ 5.7	5.8 $\pm$ 0.6
PVA/2CNF–0Li–40MMT	93.7 $\pm$ 0.5	422.3 $\pm$ 32.6	2428.5 $\pm$ 164.9	182.9 $\pm$ 19.2	284.4 $\pm$ 6.1	45.8 $\pm$ 1.9
PVA/2CNF–10Li–30MMT	94.2 $\pm$ 0.9	455.9 $\pm$ 41.7	1555.9 $\pm$ 109.4	125.9 $\pm$ 12.7	267.3 $\pm$ 5.9	42.7 $\pm$ 2.1
PVA/2CNF–20Li–20MMT	93.7 $\pm$ 1.6	456.4 $\pm$ 10.7	1237.1 $\pm$ 106.7	101.8 $\pm$ 10.6	264.4 $\pm$ 6.5	36.7 $\pm$ 1.4
PVA/2CNF–30Li–10MMT	93.9 $\pm$ 0.7	465 $\pm$ 62.8	726.1 $\pm$ 65.2	64.2 $\pm$ 6.4	260.9 $\pm$ 7.1	34.1 $\pm$ 1.2
PVA/2CNF–40Li–0MMT	93.7 $\pm$ 0.3	482.8 $\pm$ 56.9	560.6 $\pm$ 49.3	54.1 $\pm$ 4.9	257.3 $\pm$ 1.9	32.3 $\pm$ 0.9

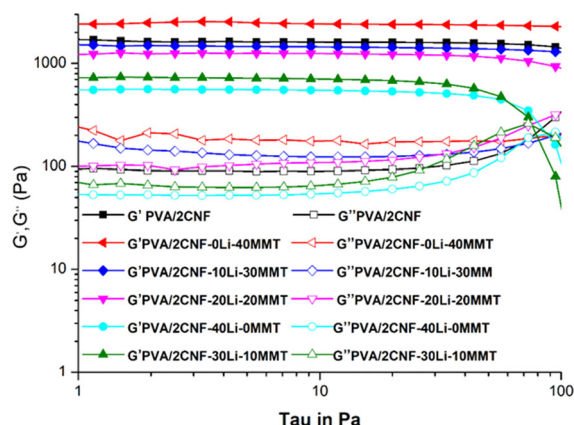


Fig. 2 Stress sweep of shear storage modulus (G') and loss modulus (G'') of hybrid hydrogels reinforced by AL or MMT at various loadings

3615, 1615, 996, and 916 cm^{-1} of MMT corresponded to the Al–OH, –OH, and Si–O–Si stretching vibrations, respectively (Tao et al. 2020). For PVA, the peaks at 3300, 2920, 1728, 1230, 1096, and 844 cm^{-1} were due to –OH, –CH₂, –C=O, –C–H, C=O, and C=C stretching vibrations, respectively (Liu et al. 2020; Chen et al. 2017).

The CNF sample exhibited two characteristic peaks at about 16.2° and 22.6°, may correspond to the (110) and (200) reflection planes of cellulose I β structure, respectively, revealing that TEMPO oxidation and ultrasonic dispersion retained the inherent crystal structure of cellulose. The broad peak at 16.2° in Fig. 1b could be deconvoluted into two peaks at approximately 14.5° and 16.9°. However, due to presence of lignin and hemicellulose, the two distinct peaks for cellulose I β were difficult to resolve. According to the XRD diffraction pattern of AL, there were broad peaks at approximately 23°, which may not represent a crystalline structure (Park et al. 2010). The XRD spectra of MMT and PVA are shown in Fig. 1b. The spectra show the characteristic peak of MMT at 6.37°, while PVA exhibits two peaks at 19.5° and 22.7°. The AFM image suggests that the nanofibers had small width and narrow size distribution. Also, the height distribution of CNF was nearly 7.2 nm and its length was about 0.95 μm (Fig. 1c–d). Compared with the width of CNF (10.6–83.0) from different materials, the width of CNF using TEMPO oxidation from bleached dry hardwood pulp in this work was higher (Hu et al. 2016; Soni et al. 2015; Tibolla et al. 2014; Wang et al. 2018). It could be

because diameter of CNF prepared from different materials is different. When compared to conventional acid hydrolysis, TEMPO-mediated oxidation adds negative charges to the cellulose surface without causing degradation of the amorphous region. The results show that TEMPO oxidation treatment is an effective method for the preparation of CNF with high aspect ratio.

Moisture content and dynamic viscoelasticity of PCAM hydrogels

Moisture content is an important factor affecting the hydrogel properties. As shown in Table 2, the water content of the hydrogels was not significantly changed in the presence of alkali lignin or MMT. It remained at about 94% due to the interactions between hydroxyl groups in adjacent polymer chains. Besides, due to the presence of alkali lignin or MMT, PCAM appeared opaque while PVA/CNF appeared transparent.

Figure 2 presents the stress scanning measurement of viscoelasticity of PCAM hydrogels at 25 °C. The storage modulus (G') and loss modulus (G'') indicate the elasticity and viscosity of the material, respectively. For the PCAM hydrogel samples, the storage modulus (G') was higher than the loss modulus (G''). This indicates that the prepared hydrogel had typical hydrogel properties. The position and length of the linear viscoelastic region (LVR) further suggest that the sample could resist flow and stress. When the addition amounts of MMT were 0, 10, 20, 30, and 40 wt%, the storage modulus of PCAM hydrogel reached 560.6, 726.1, 1237.1, 1555.9, and 2428.5 Pa, respectively. The storage modulus increased greatly with the rise in addition amount of MMT. PVA/2CNF-20Li-20MMT, PVA/2CNF-10Li-30MMT, and PVA/2CNF-0Li-40MMT exhibited higher strength and storage modulus than those of the hybrid hydrogels (CNF:PVA = 1:4) prepared by Bian et al. (2018).

As aluminosilicate in MMT contains multiple hydroxyl groups on its surface, which is beneficial for the formation of hydrogen bonds, the viscoelasticity of the hydrogel can be improved through multiple hydrogen bond networks (Liang et al. 2016). As presented in Table 2, the addition of AL reduced viscoelasticity of the hydrogel to a greater extent than PVA/CNF, which is probably because low molecular weight of AL could reduce the interactions between polymers. A crosslinking agent was not used

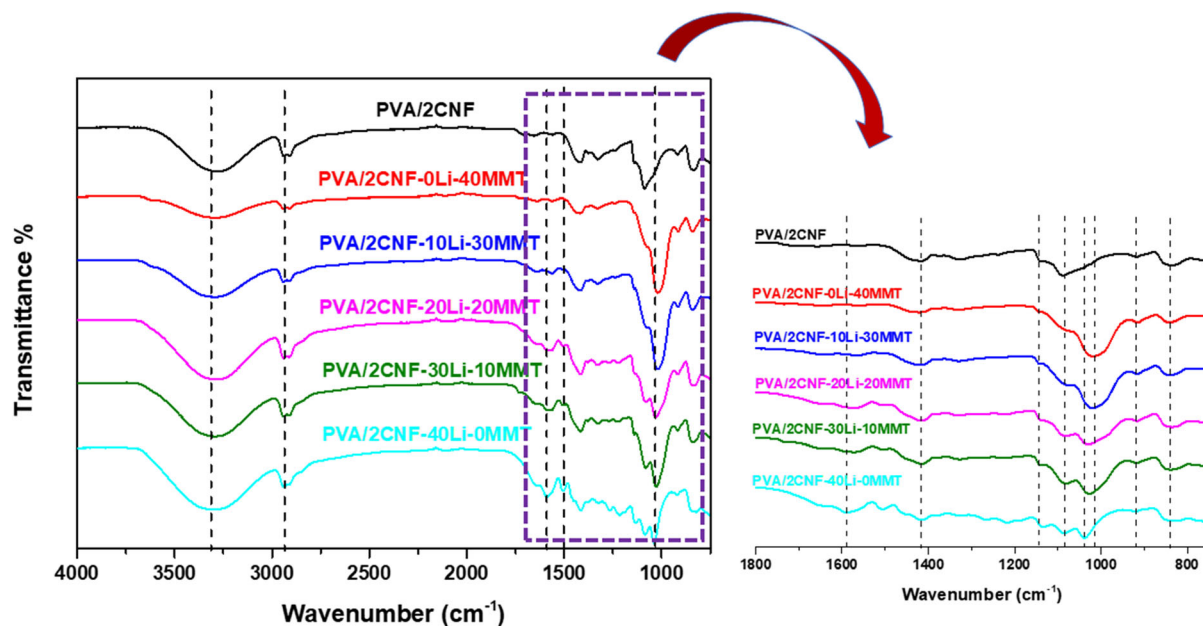


Fig. 3 FTIR spectra of hybrid hydrogels

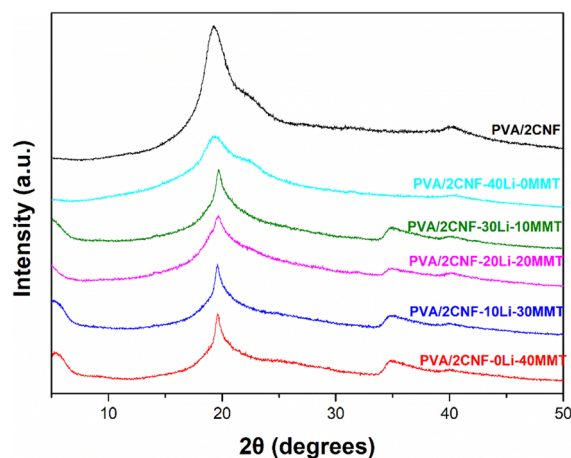


Fig. 4 XRD patterns of hybrid hydrogels

in the preparation of these hydrogels. Good mechanical properties of PCAM hydrogels containing a high content of MMT makes them good candidates for biomaterials. Also, due to strong adsorption capacity of MMT, the composite hydrogel has great potential in wastewater purification.

FTIR and XRD analyses of PCAM hydrogels

FTIR spectra (Fig. 3) of PCAM showed peaks occurring at 841, 920, 1087, and 1426 cm^{-1} corresponding

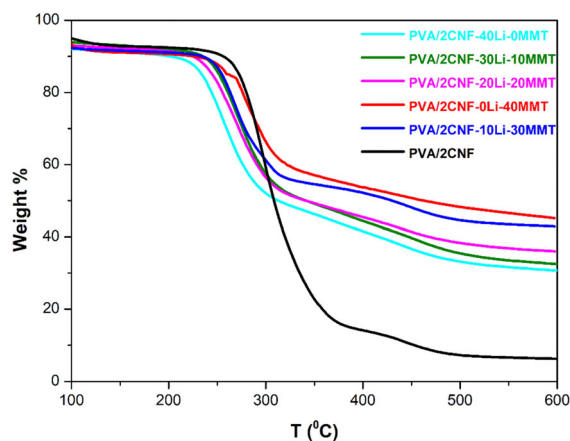


Fig. 5 TGA weight loss of hybrid hydrogels

to the stretching of C–O, bending of $-\text{CH}_2$, rocking of $-\text{CH}$, and stretching of $-\text{CH}$, respectively. FTIR spectra illustrates the presence of CNF in PCAM hydrogels, which is demonstrated by some characteristic peaks at 1034 and 1059 cm^{-1} . These peaks are related to the vibration stretch of C–OH bonds in secondary and primary alcohols of cellulose, respectively (Liu et al. 2013). Notably, compared with PVA at 1087 cm^{-1} , the peak intensity of PCAM hydrogels decreased, indicating that the crystal structure of PVA was changed due to the addition of CNF. These results

provide evidence that the interaction between the hydroxyl groups of the PVA matrix and the surface hydroxyl groups of CNF is beneficial to improve the viscoelasticity of hydrogel (Bian et al. 2018). Weak bands were observed around 1595 cm^{-1} and 1505 cm^{-1} in the PVA/2CNF-10Li-30MMT, PVA/2CNF-20Li-20MMT, PVA/2CNF-30Li-10MMT, and PVA/2CNF-40Li-0MMT samples that were not present in PVA/2CNF-0Li-40MMT and PVA/2CNF hydrogels. These bands correspond to the aromatic skeletal vibration of lignin (Schwanninger et al. 2004). The peaks at 3615 , 1615 , 996 , and 916 cm^{-1} correspond to the stretching vibrations of Al–OH, –OH, and Si–O–Si in PVA/2CNF-0Li-40MMT, PVA/2CNF-10Li-30MMT, and PVA/2CNF-20Li-20MMT. Owing to a small amount of MMT in PVA/2CNF-30Li-10MMT, the determination of characteristic peak of MMT was difficult.

XRD studies were performed on PCAM hydrogels to explore the changes in their crystallinity. XRD diffraction pattern of the hybrid film is displayed in Fig. 4. Pure PVA hydrogel exhibited a diffraction peak at $2\theta = 19.5^\circ$, which is the peak corresponding to the orthorhombic lattice structure of semi-crystalline PVA. Incorporation of CNF into the PVA system led to the formation of some new diffraction peaks located at approximately 16.6° , 22.6° , and 34.5° , which are typical reflection planes of cellulose I. However, these diffraction peaks were not prominent due to the low

content of CNF. Notably, the diffraction peak of PVA decreased with the rise in content of lignin or MMT. This phenomenon indicates that the hydroxyl radical in PVA had a strong interaction with AL and MMT, and their complexation may have destroyed the PVA network structure. Considering the PCAM composites, the intensity of the characteristic peak of MMT at 6.3° seemed to decrease. The peak of PCAM hydrogels containing MMT (hydrogel wherein MMT was added to PVA/CNF matrix) slightly shifted to left (from 6.3° to 5.1°) compared to that of MMT. This phenomenon shows that the height of MMT was probably increased due to enhanced hydrogen bond interactions. This result is consistent with the FTIR result. This result also confirms that during the preparation of PCAM hydrogels, PVA/CNF was intercalated into the interlayer of MMT nanosheets to form a composite hydrogel (Chen et al. 2018).

Thermal stability of PCAM hydrogels

The thermal stability and degradation behavior of materials are critical in biocomposite processing. Figure 5 shows the thermogravimetric analysis of hybrid hydrogels. $T_{\max}$ (the maximum weight loss temperatures of nanocomposites) and the residual mass of materials at 600°C are summarized in Table 2. Among all the samples, PVA/2CNF demonstrated highest onset degradation temperature of

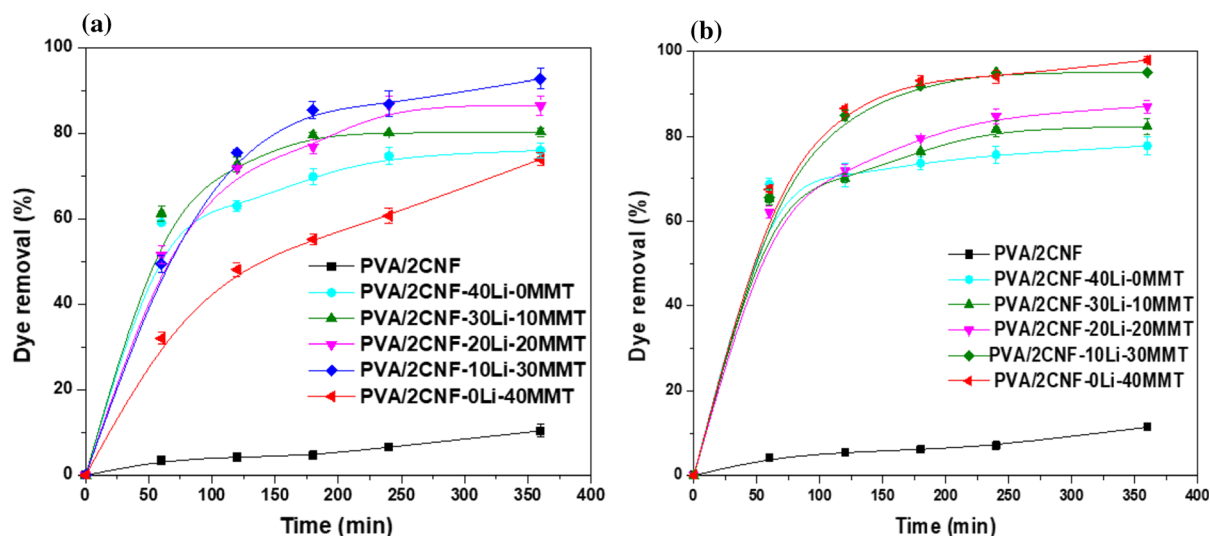


Fig. 6 Effect of contact time on removal efficiency (R%) of hybrid hydrogel samples for 250 mg/L of MB under the conditions of 30 °C (a) and 50 °C (b)

287.6 °C. However, as the AL content increased, $T_{\max}$ of PCAM hydrogels decreased from 284.4 °C to 257.3 °C. This result suggests that the PCAM hydrogels had low thermal stability primarily due to the presence of small molecular weight AL. Notably, the residual mass of hybrid hydrogel containing MMT was higher than that of PVA/2CNF hydrogel (residual mass was 5.8 wt% at 600 °C). Thermal stability of PVA/2CNF could be significantly improved due to the addition of MMT. This is mainly because MMT is an inorganic material that can hinder oxygen transport. In other words, enhanced thermal stability of the film was due to the brick mortar structure which can enhance the diffusion path of combustion gases and reduce the diffusion of oxygen in the film (Triantafyllidis et al. 2006).

Adsorption performance of PCAM hydrogels

The adsorption efficiency of 30 mg of different adsorbents was evaluated for the adsorption of 250 mg/L of methylene blue dye (MB). Figures 6a–b shows the comparison of removal efficiency of composite materials in the removal of MB from water at different temperatures. With the rise in temperature, the removal rate of MB by hydrogels was not significantly increased. When the addition amount of MMT was 40 wt%, the removal efficiency of the hydrogel at 50 °C increased significantly, which may be because due to the presence of MMT, the cross-linked structure of the PVA/CNF network became relatively dense, affecting the MB adsorption at low temperature. This result is consistent with the conclusion in Fig. 2. In the case of adsorption by PVA/2CNF-0Li-40MMT sample for 360 min at 50 °C, the MB removal rate reached 92.7 wt%. The adsorption performance decreased with the increase in mass ratio of AL. When the AL content in PCAM hydrogels reached 40 wt%, the MB removal rate was only 77.8 wt%. This may be because the affinity of binding sites of MB in AL is lower than that of MMT, which is not conducive for the adsorption of MB. As per the results of removal rate of MB at different temperatures, a higher AL content loosens the cross-linked network of the hydrogel, so that the hydrogel can bind to MB more quickly. MMT is a phyllosilicate clay and it has a sandwich structure consisting of two tetrahedron layers that are sandwiched between the edge-bridged octahedral sheets. It has a three-dimensional porous

structure and a large specific surface area; thus, having high adsorption performance. Higher the MMT amount, greater was the adsorption capacity. The absorption capacities of the PCAM hydrogels were one order greater than those of the natural sorbents and 3–6 times superior to the PVA/Chitin/CNF composites (Mok et al. 2020; Ng and Leo 2019). As mentioned in other works, high amounts of MMT can be incorporated in PCAM hydrogels, which reduced the swelling ratio of the hydrogel (Sirousazar et al. 2011). For the hydrogel samples, a higher swelling ratio signifies that the network density is lower. Herein, slightly lower swelling ratio for PVA/2CNF-10Li-30MMT was observed as compared to PVA/2CNF-20Li-20MMT, which is again lower than the swelling ratio of PVA/2CNF (Table 2). These results are in agreement with the dynamic viscoelasticity results that suggest the same order in terms of cross-linking density.

Sorption kinetics of PCAM hydrogels

To study the MB adsorption rate of PCAM hydrogels at 50 °C, the linear pseudo-first- and second-order kinetic equations were employed. R^2 values can determine whether the system is related to the models.

The pseudo-first-order equation is shown in Eq. (6)

$$\ln(q_e - q_t) = \ln q_e - \frac{k_1}{t} \quad (6)$$

The pseudo-second-order equation is shown in Eq. (7)

$$\frac{t}{q} = \frac{1}{k_2 q_e^2} + \frac{t}{q_e} \quad (7)$$

where q_t is the adsorption amount (mg/g) of MB at time t (min), q_e is the equilibrium adsorption amount of MB (mg/g), and k_1 and k_2 are the pseudo-first- and pseudo-second-order adsorption rate constants (g/mg·min), respectively.

As shown in Fig. 7a–b, the correlation coefficient (R^2) of pseudo-first-order kinetic equation was lower than that of the pseudo-second-order kinetic equation. Also, the correlation coefficient (R^2) of pseudo-second-order kinetics of MB adsorption by the hydrogel containing AL or MMT was greater than 0.991. The calculated q_e value was similar to the experimental q_e value. the MB adsorption by PCAM hydrogels at 50 °C was in accordance with the pseudo-second-order kinetic model (Kumar et al. 2008).

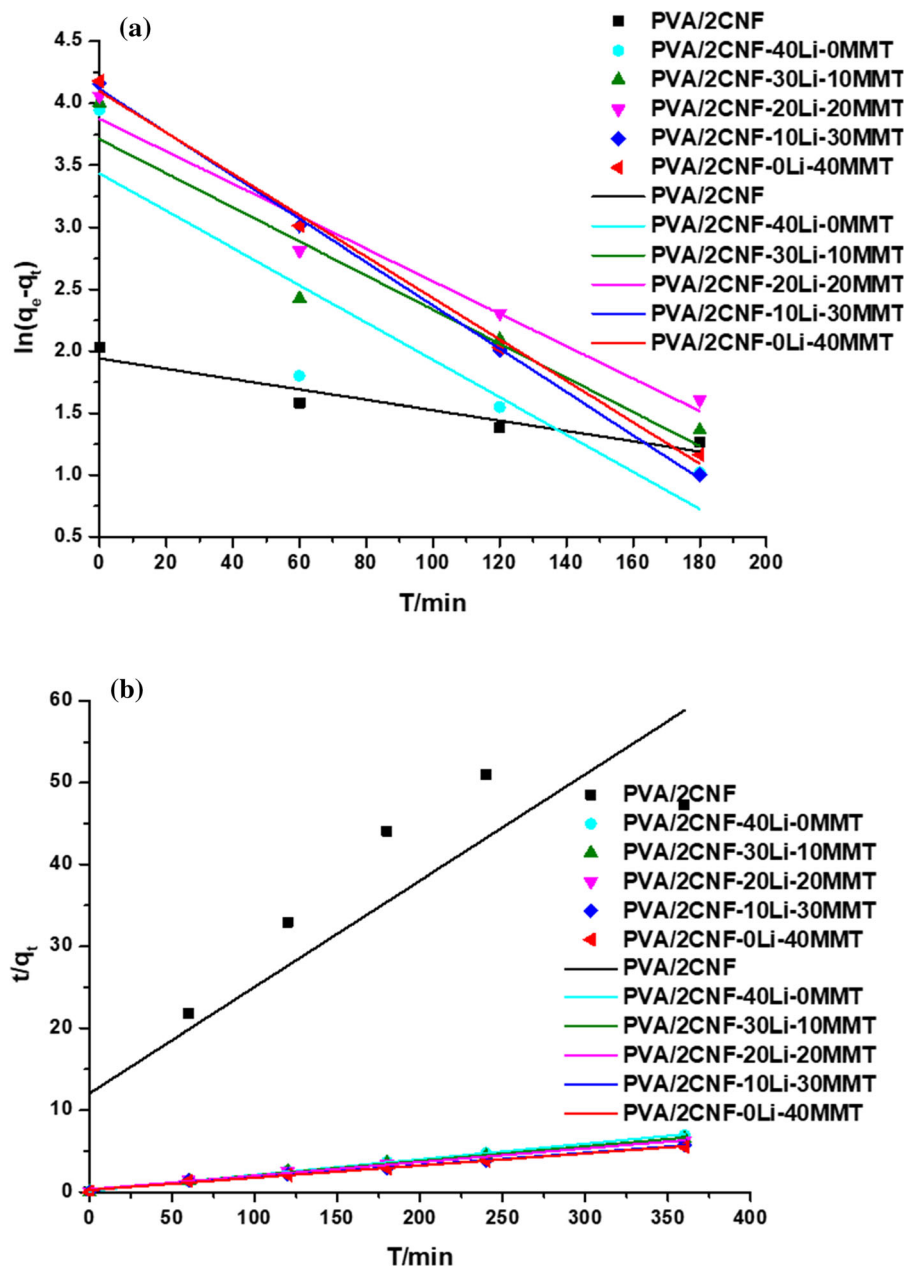


Fig. 7 Pseudo-first-order (a) and second-order kinetics (b) of MB by hydrogel samples at 50 °C

By increasing pH from 2.4 to 10.5, the dye removal (%) of PVA/2CNF-10Li-30MMT rose from 85.8% to about 94.2% (Fig. 8). The relatively high % of dye removal at pH 10.5 could be because there was an electrostatic repulsion between similar ionic charges owing to the presence of H^+ in the acidic medium and MB cations. Therefore, the decrease in pH hindered

the MB adsorption on PCAM hydrogels (Sadik et al. 2020) (Fig. 9).

Herein, the adsorption/desorption experiment was repeated five times (Fig. 8). Specifically, the PVA/2CNF-10Li-30MMT that adsorbed MB was eluted with acetic acid and ethanol ($v/v = 1/9$) solution. The removal efficiency of MB was 92.7% in the first cycle and still 33.6% by the fifth cycle. Thus, the results

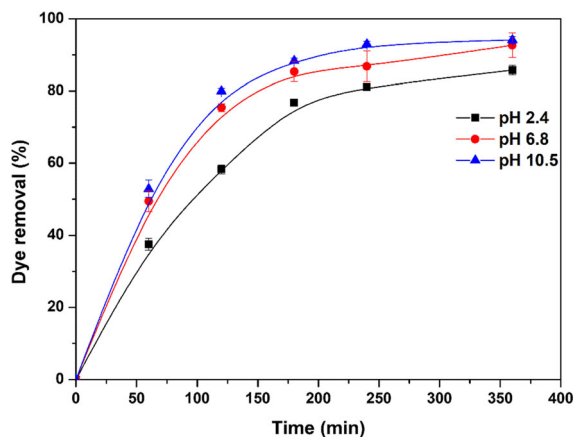


Fig. 8 Effect of pH on dye removal (%) of MB adsorbed onto PVA/2CNF-10Li-30MMT

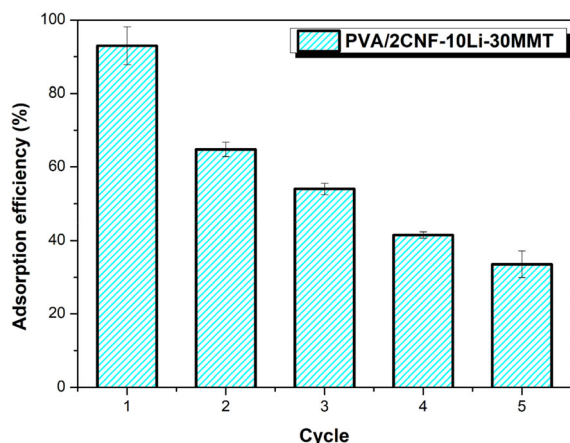


Fig. 9 Adsorption amount for phosphate in five adsorption-desorption cycles of PVA/2CNF-10Li-30MMT

indicate the reusability of PCAM hydrogels. From environmental and economic assessment considerations, regeneration and recyclability are beneficial for the adsorbent material.

Conclusions

PVA hydrogel with good adsorption and mechanical properties was successfully prepared without the addition of initiators or chemicals. By the incorporation of CNF, AL, and MMT into PVA water-based system, the viscoelasticity of composite hydrogel changed. The hydrogels with high AL content had lower viscoelasticity than those with large

montmorillonite content. MMT was well-dispersed in the polymer matrix, acting as a multifunctional crosslinking agent to enhance interactions between MMT and PVA. However, an excessively high MMT content slowed down the adsorption efficiency. The removal rate of MB by PVA/2CNF-10Li-30MMT reached 91.8% in 120 min. Hence, addition of a small amount of lignin effectively improved the adsorption efficiency of hydrogel. The results from absorption tests also suggest that the removal of MB followed the Lagergren's quasi-two-stage adsorption kinetic model and the adsorption process proceeded on a non-uniform surface. To conclude, this research has practical importance in the improvement of utilization of lignocellulosic materials in various applications.

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Author's contributions JL designed the project, performed experiments, analyzed data, and prepared the manuscript. XM and XZ helped to analyze the data and revised the manuscript. YX supervised the project and revised the manuscript. All authors read and approve the final manuscript.

Availability of data and materials The datasets used and analyzed during this article are availability from the corresponding author on reasonable requests.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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